Microgrid demand forecasting on Zoho Catalyst

GridPilot

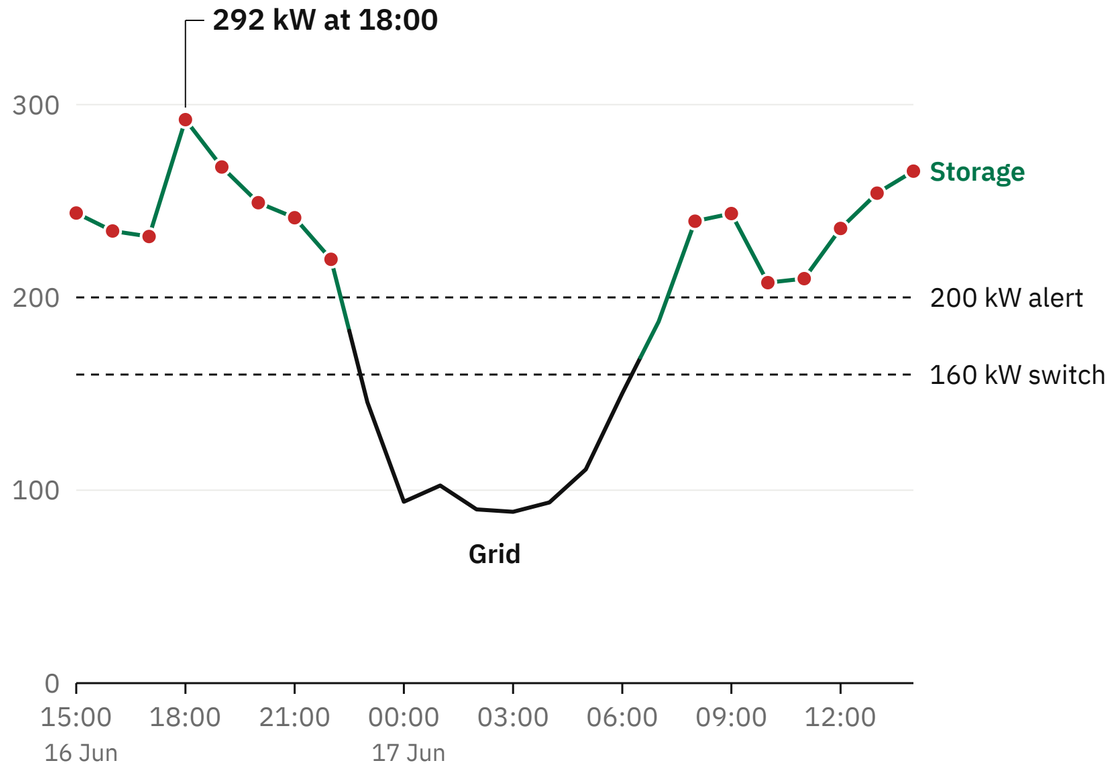
Forecast tomorrow’s load.
Switch before it spikes.

Team QUANTIX · HackSphere 3.0, BITS Pilani Hyderabad

Yashvi Chotalia (lead), Shubham Sahu, Aditya Jadhav, Samhita Hejmadi

Live app
`gridpilot-50047251149.development.catalystappsail.in`

Predicted demand, next 24 h kW



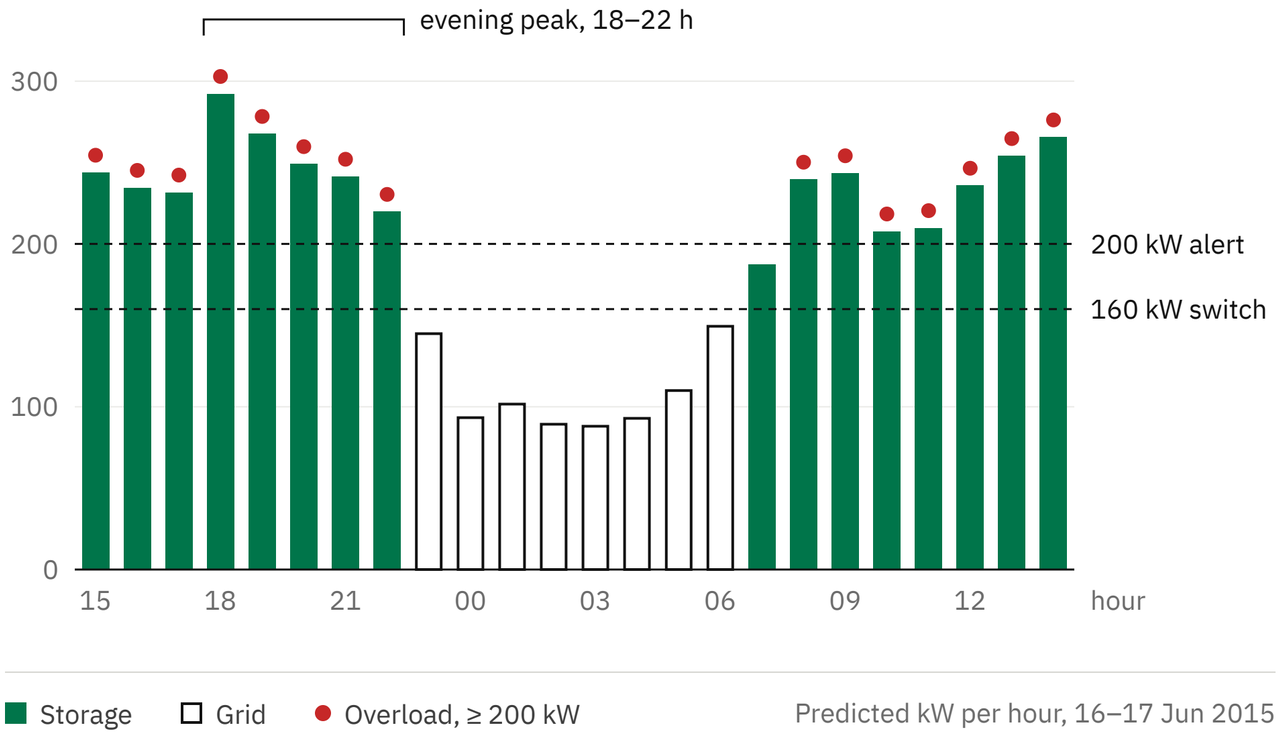
Live app output, 16–17 Jun 2015 sample day

● Overload hour, ≥ 200 kW

Source: GridPilot live app, `/api/forecast`

02 Problem

Every hour a microgrid must pick grid or battery, and the test weeks run hotter than most of training.



24

source decisions per day, one every hour

15 of 24

hours at or above 200 kW in the sample day

236 of 1,000

test hours above the training 95th percentile, 32.6 °C

Source: live app forecast; `train_telemetry.csv`; `sample_test_input.csv`

03 Solution

Weather goes in. A 24-hour switching plan comes out.

1 **Ingest**
Form or CSV upload

2 **Clean**
Validate, impute, clip

3 **Forecast**
QuickML + hybrid model, 24 h

4 **Decide**
Transition controller rules

5 **Act**
Dashboard, alerts, dispatch signals



The live dashboard

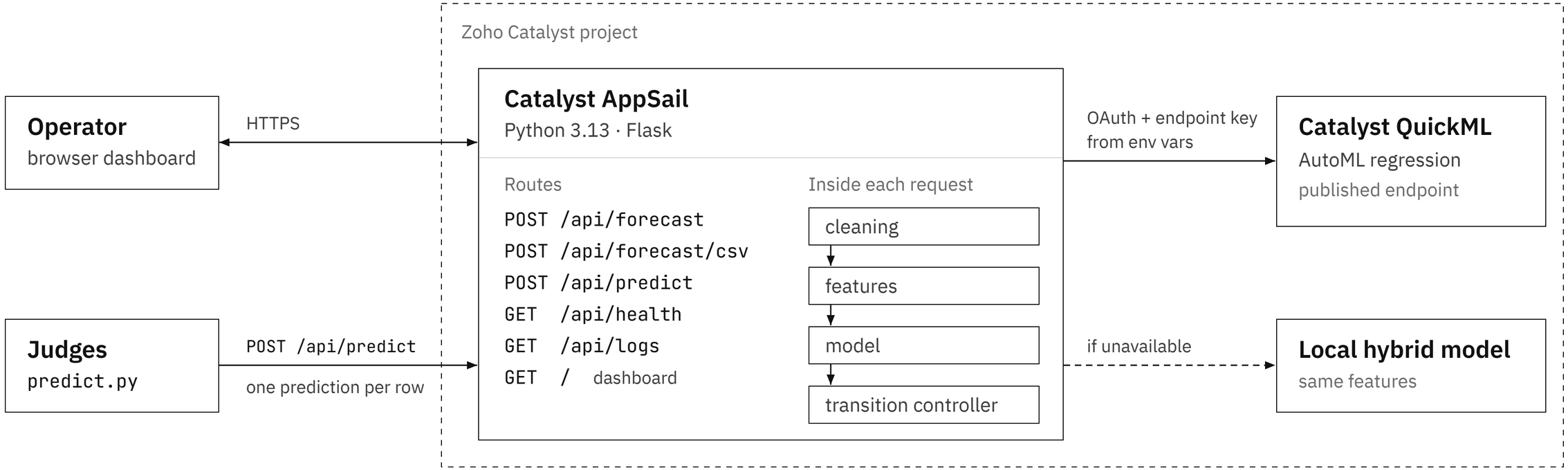
After one forecast run: the overload and battery-reserve alerts, the active source for the hour, and the rule that chose it.

Open it
`gridpilot-50047251149.development.catalystappsail.in`

Source: GridPilot live app on Zoho Catalyst AppSail

04 Architecture

Everything runs on Zoho Catalyst: one AppSail service and one QuickML endpoint.



Secrets live in environment variables, never in git.

Deployed with the Catalyst CLI.

Source: hacksphere-gridpilot repository, `appsail/`

05 Data

4,000 training hours, cleaned the same way at training and prediction time.

Inputs, one row per hour

timestamp	
ambient_temp_c	°C
humidity_pct	%
solar_irradiance_wm2	W/m²
Target	
target_demand_kw	kW

4,000 rows, Jan to Jun 2015

Messy upload, after cleaning

timestamp	temp °C	humidity %	W/m²	fix
2015-06-17 15:00:00	35.7	56.4	169	none
2015-06-17 16:00:00	▸ 35.0 was blank	54.7	112	imputed
2015-06-17 17:00:00	34.3	▸ 100 was 140	52	clipped
▸ 2015-06-17 18:00:00 was 17-06-2015 18:00	35.4	56.9	▸ 0 was -5	date normalised, clipped
~~not-a-date~~	~~32.1~~	~~56.4~~	~~2~~	**dropped**
2015-06-17 20:00:00	30.0	59.1	0	none

Example rows. The counts on the right are from a real upload to the live app.

Real upload to the live app

6	5
rows in	rows out

Dropped	1
Imputed	1
Clipped	2

`POST /api/forecast/csv`

Features: hour-of-day profile, weekday or weekend, cyclic hour (sin, cos), cooling degree (T – 22 °C), heating degree (15 °C – T), humidity, solar irradiance, evening-peak flag 18–22 h.

Source: live `POST /api/forecast/csv` test, 28 Oct 2026

06 Model

On weeks it never saw, the hybrid model scores 9.70 kW RMSE, five times better than the baseline.

Hybrid model, RMSE on the test weeks

9.70 kW

MAE **7.74 kW**

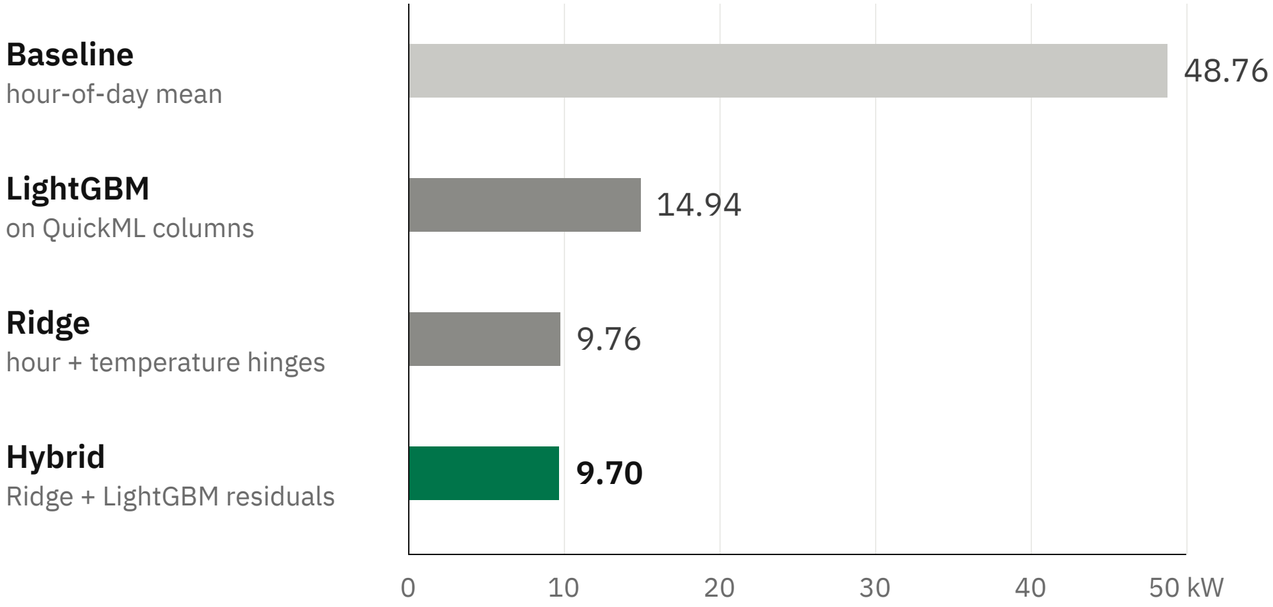
Chronological split, no shuffling



Train, 80%
Jan to 14 May 2015

Test, 20%
14 May to 16 Jun

RMSE on the test weeks, kW. Lower is better.

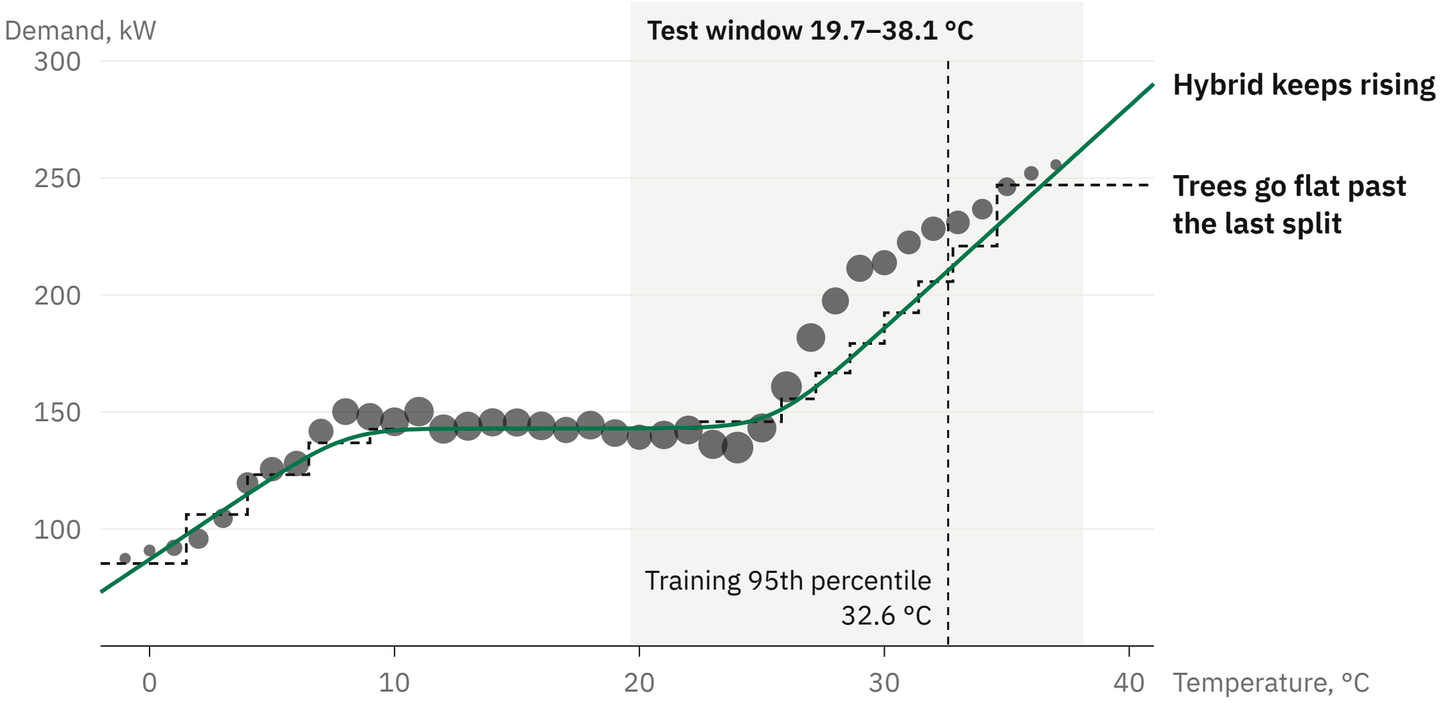


The ridge base keeps rising with temperature; small boosted trees correct what is left.

Source: `model/RESULTS.md`, chronological hold-out, no shuffling

07 Why hybrid

The test weeks run hotter than 95% of training, so the model has to extrapolate.



Dots: real training data. Lines: illustration of how each model type behaves.

236 of 1,000

test hours are hotter than 95% of the training data

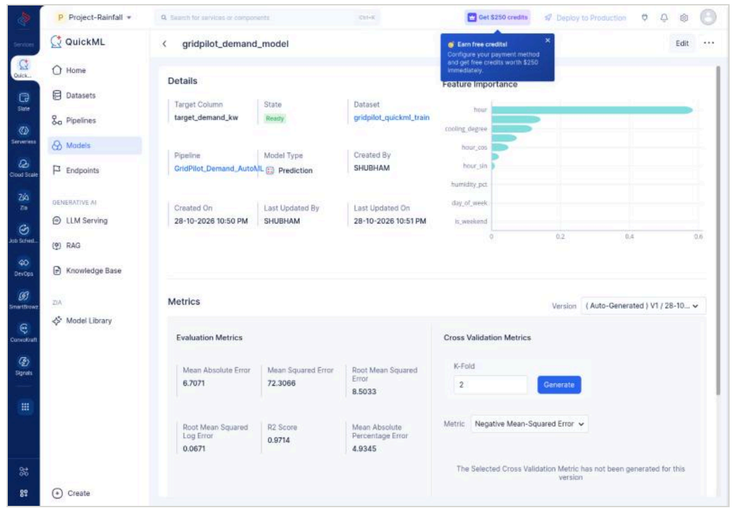
Training 95th percentile	**32.6 °C**
Test window	**19.7–38.1 °C**

08 QuickML

We trained and published the model in Catalyst QuickML: R² 0.971 on its evaluation split.

1. **Dataset**
   `gridpilot_quickml_train`
   4,000 rows · 14 columns · quality 10/10
2. **Mean imputer**
3. **Min-max normalisation**
4. **Gradient-boosted regression**
5. **Published endpoint**
   `gridpilot_demand_endpoint`

R²	RMSE	MAE	MAPE
0.971	8.50 kW	6.71 kW	4.93%



QuickML model page, Catalyst console

Top features

1. `hour`
2. `cooling_degree`
3. `hour_cos`

`POST …/quickml/v1/project/…/endpoints/predict → HTTP 200, 197.89 kW`

Source: Zoho Catalyst QuickML console, 28 Oct 2026

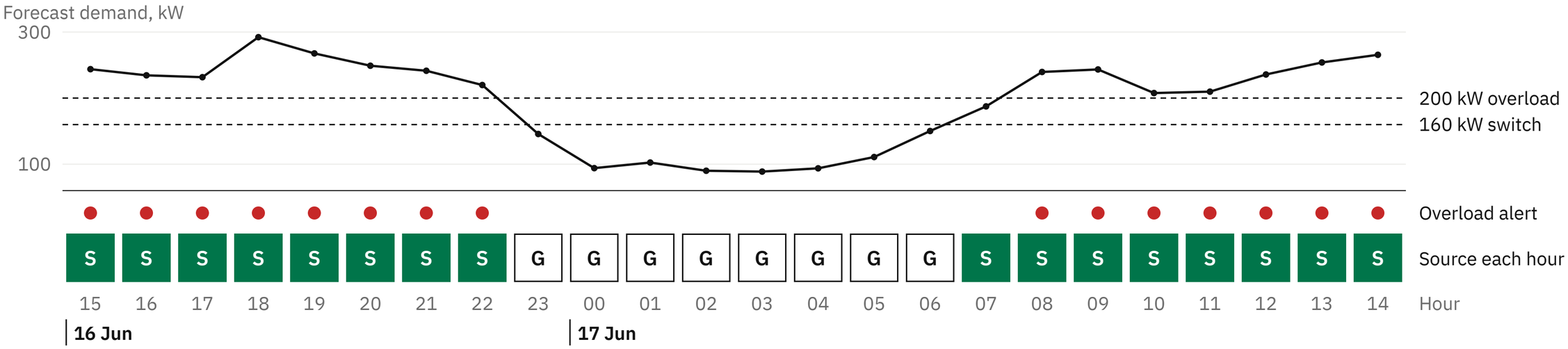
09 Controller

Three rules decide the power source, and every boundary is tested.

Storage
Demand > 160 kW, or hour 18:00–22:00 inclusive

Grid
Otherwise (≤ 160 kW and outside 18–22)

Overload alert
Demand ≥ 200 kW, independent of mode



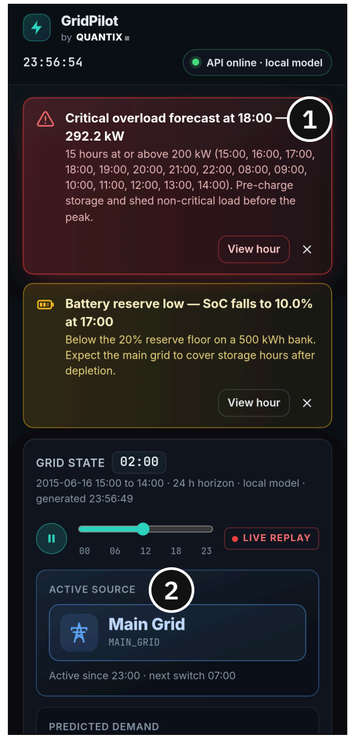
160.00 kW → Grid	**160.01 kW** → Storage	**200.00 kW** → Alert	**22:00** → Storage	**23:00** → Grid	**9 boundary tests pass**

Source: live app `/api/forecast`; `tests/test_controller.py`

10 Dashboard

Operators see the next 24 hours, the active source and every alert on one screen.





Same screen on a phone

1. Overload alert with hour and kW
2. Active source and battery charge
3. 24-hour dispatch timeline
4. Daily figures
5. CSV and manual input
   Further down the page

Peak **292 kW** at 18:00 | **16 h** storage / **8 h** grid | **15** overload hours | **2** switches

Source: GridPilot live app, sample day 16–17 Jun 2015

11 Judging criteria

How GridPilot answers each judging criterion.

Criterion	Weight	What we built	Evidence
Model accuracy QuickML	**35%**	Hybrid model + QuickML AutoML model	RMSE 9.70 kW on chronological hold-out; QuickML R^2 0.971
Deployment & functionality	**35%**	Live AppSail app, CSV + manual input, public API	`test_eval.py` → SUCCESS from a fresh clone
Transition controller	**20%**	3 rules, 160/200 kW, 18–22 h	9 boundary tests pass
UI & UX	**10%**	Dashboard with alerts, timeline, battery	Works on desktop and phone

```
$ python test_eval.py
SUCCESS
$ python predict.py --input sample_test_input.csv --output predictions.csv
Wrote 1000 predictions
```

Run from a fresh clone of `hacksphere-gridpilot`

24 tests pass

15 API tests and 9 controller boundary tests

Source: HackSphere 3.0 problem statement, section 7; hacksphere-gridpilot tests

12 Team

Team QUANTIX

Yashvi Chotalia	Team leader
Shubham Sahu	
Aditya Jadhav	
Samhita Hejmadi	

Links

Live app	`gridpilot-50047251149.development.catalystappsail.in`
Code	`github.com/https-shubhamsahu/hacksphere-gridpilot`
Leader GitHub	`github.com/YashviChotalia`

Scan to open the live app

Next steps

Questions welcome.

1. Live meter feeds via Catalyst Event Listeners
2. Forecast history in Catalyst Data Store
3. Per-site models through QuickML pipelines
4. SMS alerts before the overload hour